



A Number You Can Derive

WisdomForge Booklet — Ages 15–18

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About This Book

If you cannot walk the path on a blank board, you are not ready to turn it in — and you are not ready to drive a system that depends on the number.

This is the high band of *Number in the Age of AI*. The little sittings already exist: guess, path, catch, two ways, about, machine. Little 6/6 is live. Young sitting 1 (‘estimate-first’) already exists. This booklet does not invent sittings. It treats the same six beats as academic integrity.

The Hour books (*Mathematics for Homeschooling*, the banded Math Hours) are a different product. Do not skip this spine because those exist. Critical Thinking’s four-band Hour set is already 200; this is not that product either.

You will meet real arguments, not slogans. George Pólya’s *How to Solve It* (1945) is the method book: unknown, data, condition; understand, plan, carry out, look back. Richard Feynman’s 1974 Caltech address is the honesty book: do not fool yourself. Richard Skemp (1976) named instrumental versus relational understanding. John Sweller’s worked-example research says novices learn from examples — then have to walk. Gerd Gigerenzer’s numeracy work is the civic twin: size and base rate before a fluent percent. Fermi estimates are a public chain you can argue with, not a costume census.

CAS, Photomath, and chat models are in scope. Accommodations are access, not a ghost. Medical dosing is a professional, not a model. One chapter ends in a Research Prompt. The other five end in Reflect. Practice items are protocols with a deliverable.

Chapter 1: Ballpark Before the Oracle



Illustration for Chapter 1

A model will emit digits with great confidence. If you do not know whether the answer should be near 4, 40, or 400, you will accept 47 and move on. Estimation is how you keep the oracle honest. It is also how you keep a headline, a budget line, and a lab result honest.

Write the ballpark with a unit. Show the steps. Then a tool may audit the steps. If it only offers the digit, refuse. Reverse-engineering a ballpark after seeing the answer is theater. The ballpark has to exist before the tool, or it is not a ballpark. It is a shadow.

Pólya starts where every honest solution starts: What is the unknown? What are the data? What is the condition? The unknown is not “whatever the solver prints.” The unknown is a quantity in the world — time, money, count, length, rate — with a unit. A fluent 47 with the wrong unit is a miss. Forty-seven minutes is not forty-seven miles. The costume got more fabric. The trick is the same as little’s grapes.

Fermi’s order-of-magnitude estimates belong here as a *public chain*. How many piano tuners in a city? You do not need the census first. You need a sequence of abouts that could be wrong in public: population, fraction of households with pianos, tuning frequency, tuners’ workload. Wrong in public is how you get a sense of size. A chatbot that prints 1,247 tuners with no chain has given you a costume. You cannot argue with a costume. You can argue with a chain.

There is a real scholarly objection to Fermi practice: that rough chains license sloppy policy, that “about” becomes a way to skip measurement. Take the objection at full strength. A Fermi chain is not a substitute for the Bureau of Labor Statistics. It is a prior. Priors exist so you can notice when a later number cannot fit. If you treat the prior as the finding, you have cargo-culted Fermi. Feynman would recognize the imitation of a form — the napkin arithmetic — without the honesty of saying what you did not measure.

Gigerenzer’s point is civic. People who cannot name a size will believe a fluent percent. “47%” with no base is not information. “About one in two of a group you can name” is information you can argue with. Headlines prefer the costume. This chapter prefers the size. NCTM has treated estimation as a standard, not a warm-up game, for decades. If a pacing guide skips it because a calculator is faster, the sitting still runs. The calculator can wait. So can the model.

The emerging sitting *Show the Path* already said the adult version of the same law: a number you cannot derive is a rumor you are willing to sign. High school is where that rumor starts to have transcripts attached. College will not care that a chatbot can integrate. An exam hall will. So will a lab, a newsroom, and anyone who has to stake money or safety on a digit.

If they say the calculator is the method, the sitting’s reply still holds: the calculator is an instrument. The method is the sentences you could say without it. If the method is not on the page, the digit is not yours.

A worked Fermi you can actually run: a gym. You have stood in it. Rows of bleachers, a court, a fire-code sign if one exists. About how many bodies before it feels dangerous? Write the chain: rows $\times$ people per row $\times$ sides, plus floor. The largest uncertainty is packing. A model that prints 2,847 without asking what you saw has not estimated your gym. It has emitted a gym-shaped number. If a later official capacity is 1,200 or 4,000, your chain either held or it taught you which link was weak. That lesson does not come from the printed digit. It comes from the link you can point at.

Order of magnitude as civic skill is not a metaphor. A budget story that moves a decimal is a unit error with consequences. A campaign ad that treats “millions” and “billions” as synonyms is counting on you not having a ballpark. You do not need to become an economist tonight. You need “thousand / million / billion” as three different rooms, the way little needed “about ten” versus “forty-seven.” The grapes scaled up. The alarm is the same muscle.

Pólya’s four phases, mapped without costume: understanding is the ballpark with unit; planning is the method on the page; carrying out is the count or the algebra, still yours; looking back is the catch, the second path, the honest about, the tool as reviewer. You can fail any phase by skipping to a solver. The solver will happily complete all four in a paragraph that looks like work. Looking like work is the cargo cult. Feynman was talking to scientists who built antennas that could not talk to the sky. You are being asked not to build homework that cannot talk to a blank board.

Young sitting language still applies: pick a real quantity; TRY FIRST; no live timed test as the demo if it tempts cheating; units matter. Paste the method, not “solve this.” Evaluate: estimate present? Did they refuse a bare digit? Integrity: if the method is not on the page, the digit is not yours. Dinner: what number did we believe this week without a ballpark?

The Big Idea

Ballpark first. Method second. Tool third. Digit last. A number without a size is a rumor. A Fermi chain is a prior, not a finding. Extra digits are not a measurement.

Practice

1. **Unit sheet (12 min).** Take one problem from your actual math and one household quantity (trip, receipt, battery). Write UNKNOWN / DATA / CONDITION / BALLPARK WITH UNIT / METHOD. Tool closed. Deliverable: one page. 2. **Fermi chain (15 min).** Estimate one public quantity (piano tuners is allowed; a local one is better: seats in the gym, gallons in the water tower if you can see it). Write the chain and the largest uncertainty. Do not look up the “answer” first. Deliverable: the chain, not a digit. 3. **Refuse a bare solve (8 min).** Paste your method into a model. Ask only: where is this shaky? If it returns a total, close it and note the miss. Deliverable: the question you asked and whether the tool obeyed. 4.

Headline size (10 min). One news number from this week. Write the order of magnitude and the missing base or unit, if any. Deliverable: SAID / DIDN'T SAY.

Reflect

When did you last reverse-engineer a guess after seeing an answer — and what would have been different if the ballpark had been forced to go first?

Chapter 2: Path Over Prize



Illustration for Chapter 2

Mathematics is a path other people can walk. The prize at the end is a check on the path, not a substitute for it. A model that jumps to the prize has stolen the discipline. Use it to ask “is this step legal?” after the path exists. If you cannot reproduce the path on a blank board, you are not ready to turn it in.

Little called this marks on paper. You can write sentences: I divided because the problem asked for equal groups; I converted minutes to hours because the rate was per hour. Those sentences are the path. A Photomath screenshot is not a path you can walk in an exam hall. It is a prize with a camera.

Richard Skemp, writing in 1976 for mathematics teachers, distinguished *instrumental* understanding (rules without reasons) from *relational* understanding (knowing what to do and why). A camera that boxes an x is instrumental in the extreme: you may not even hold the rule. This booklet is not anti-rule. Algorithms are paths you can walk. It is anti-prize-without-path. Instrumental knowing feels fast. Relational knowing survives a blank board. Pólya's questions are a relational drill. Photomath-first is an instrumental drill that pretends to be help. If you can only say "the app did it," you have neither.

There is a live debate here, and you should hear the other side. Some teachers argue that CAS and solvers are what professionals use, so withholding them is a museum pedagogy. The reply is not Luddism. Professionals still have to know when the instrument is insane. Pilots have instruments and still look out the window. Engineers sign calculations they can defend. The exam-hall test is a reproducibility condition, not cruelty: voice in English is what remains when the chat is closed; path in math is what remains when Photomath is closed. If "everyone will use it in college anyway," then the ones who can still walk the path will be the ones worth trusting. Be those.

Sweller's worked-example research is often misread as permission to skip doing one. Novices *do* learn from worked examples. A worked example you cannot reconstruct is a video of someone else walking. Watch, then walk. The sitting is the walk. If your homework is five camera prizes and zero reconstructed steps, you optimized for a grade costume.

CAS, Photomath, a sibling who already finished, a parent who "just wants it done" — same rule. Accommodations are different: a reader or extra time is access, not a ghost. Name access as access. Do not use access as cover for a missing path. Integrity: cite the tool if it changed a step. If you cannot cite it, you should not have used it. That is the same rule English uses for ghostwriting. Math is not exempt because the symbols look objective.

The emerging sitting said it without decoration: if you cannot walk another person through the steps, you did not do the math. A model walking them is not you. Ghostwriter ethics is not only English.

Skemp's 1976 paper is short enough to read. Instrumental understanding: rules without reasons — "to divide fractions, turn it upside down and multiply," with no picture of why. Relational understanding: a web of connections that lets you reconstruct a rule if you forget it. Solvers offer a third category Skemp did not have to name: *outsourced* understanding,

where even the rule lives in another machine. Instrumental at least lives in you as a procedure. Outsourced lives nowhere you can be examined. A student who is instrumental can still sit a paper. A student who is outsourced cannot. That is not a moral insult. It is a description of where the knowledge is stored.

The museum-pedagogy objection deserves a second paragraph. Yes, working mathematicians use CAS. Yes, data scientists use libraries they did not write. The professional move is still: I can defend this step; I know the failure modes; I can tell when the output is insane. A junior who cannot do that is not “using modern tools.” They are a liability with a login. High school is not pretending you are already a professional. It is building the defense-of-step muscle so that later tools do not eat the person.

Pólya again, as protocol not poster. Look at the unknown. Think of a familiar problem with the same unknown. Here is a problem related to yours and solved before. Could you use it? Could you use its result? Could you use its method? A solver that prints the boxed number has answered none of those for you. Sweller would say: study a worked example, then do a near one with the example closed. If the example never closes, you did not study. You watched.

Blank-board protocol from the emerging sitting, kept: one problem at the edge of current skill; paper; no tool; then audit; close; reproduce one key step from memory. Dinner: where did we sign a number we could not derive? Transfer: code you cannot explain; reviewer, not solver. If they say “in the real world we use tools,” keep the sitting’s reply: in the real world someone still has to know when the tool is insane.

Cite. If the app changed a step, write that down. Application essays: human draft. Lab reports: no model-written results. If you didn’t measure it, you don’t report it.

The Big Idea

A number you cannot derive is a rumor you are willing to sign. Path over prize. Relational understanding is what remains when the instrument is closed.

Practice

1. **Blank board (18 min).** One problem at the edge of current skill. Solve on paper. No tool. Then photograph the method and ask a guide only for legal-step critique. Close the tool. Reproduce one key step from memory. Deliverable: the reproduced step, compared. 2. **Citation log (10 min).** For one assignment this week, write every tool that touched a step. If you cannot cite it, you should not have used it. Deliverable: the log, including “none.” 3. **Instrument vs method (10 min).** Take a CAS or solver output you already have. Write three sentences you could say without it that make the first legal step. If you cannot, the output was a prize. Deliverable: the three sentences or an honest failure. 4. **Access named**

(5 min). If you have an accommodation, write one line that names it as access. If you do not, write one line about a classmate's access that is not a ghost. Deliverable: one named distinction.

Reflect

Where did “in the real world we use tools” hide a missing path this month — and who paid for the hiding?

Chapter 3: Catch a Fluent Wrong



Illustration for Chapter 3

A number can sound finished and still be wild. Little practiced this with a grown-up who lied on purpose about grapes. You will meet fluent wrongs that do not wink: a typeset fraction, extra digits, a green check, a model's confident paragraph.

Feynman, 1974: the first principle is that you must not fool yourself — and you are the easiest person to fool. Cargo-cult science, in that address, is the imitation of the forms of investigation without the honesty. A typeset answer is a form. A green check is a form. A model's extra digits are a form. The honesty is the ballpark you wrote when no one was looking.

Write a ballpark. Show a method. Then hear a confident wrong — from a planted error, from a homework app with a sign error, from a model you asked after you already knew. Say it cannot be, using the ballpark and the path. Do not wait for a teacher to announce the catch. If they announce it, they stole the alarm.

Units are the usual costume. 3.2 hours versus 3.2 minutes. \$4 million versus \$4 billion. 47 people versus 47 percent. The digits can be identical. The world is not. Extra digits are a second costume: 47.382 looks more true than 50 even when nobody measured to the thousandth. Gigerenzer again: precision is not the same as accuracy, and a base rate you cannot name is not a statistic you can use.

Do not shame a miss. Repeat. Wild is data. If your ballpark was 10 and the fluent wrong was 12, that is not this drill — 12 might be rounding. Use a wrong that cannot fit: 10 versus 400. The point is the bell, not a trick about 11.

Headlines will do this. So will social posts with a chart that has no axis. So will a solver that has a sign error and a green check. Green is a style. Truth is a check. “The tool wouldn't be wrong” is a hope, not a method. Tools drift. Training data drift. Voices sound sure. Your path is how you notice.

There is an objection worth taking seriously: constant suspicion is exhausting, and a student who treats every digit as a lie will not finish the course. The answer is not cynicism as a personality. The answer is a cheap alarm — a ballpark with a unit — that rings only when the world cannot fit. You do not need to distrust $7+5$. You need to distrust 47,000 people in a room you know holds hundreds.

Cargo cult, applied without stretching Feynman past what he said: the South Sea islanders, in his telling, built wooden towers that looked like air-traffic control and waited for planes that did not come. The form was right. The planes were not caused by the form. A homework set that looks like mathematics — boxed answers, extra digits, a paragraph of “therefore” — can be a wooden tower. The plane is a path you can walk when the chat is closed. If the path is not there, the look of math did not cause understanding. It imitated it.

Little sitting *Catch a Wild Number* is still the drill. A grown-up supplies one confident wrong total. You must say the catch. The device is not first. Repeat on a miss. Optional later: a tool question you already know, to see if the alarm still rings. Integrity: do not let the guide

announce the catch. High school just changes the pile: a word-problem, a chart, a headline, a solver output.

Units as costume deserve a catalog you can keep. Time: hours vs minutes vs seconds, 2.5×60 versus 2.5 treated as already minutes. Money: millions vs billions, a three-order error that still fits in a headline font. Percents: 47% of what. Rates: per what. Counts vs percentages that reuse the same digits. If you say the unit out loud, half the fluent wrongs die before they sit down.

Green-check apps are a special case because they mix authority with speed. Authority is a social costume. Speed is a cognitive one. Together they feel like truth. Your ballpark does not care how fast the check appeared. If 400 cannot fit a pile you guessed at 10, the green check is lying with a smile.

The Big Idea

Fluent is a style. Truth is a check. Without a ballpark, a fluent wrong number walks in. Extra digits are often a costume of measurement.

Practice

1. **Planted wrong (12 min).** Ballpark and method on a real problem, tool closed. A grown-up or a later known-wrong supplies one confident wrong with a unit. Refuse it out loud, citing ballpark and path. Then find the leak — in you or in it. Deliverable: the refusal sentence. 2. **Digit costume (10 min).** Find one number this week with extra digits and no measurement story. Rewrite it as an honest about. Deliverable: before/after. 3. **Chart without an axis (10 min).** Screenshot or describe one. Write what would have to be true for the implied size to fit a world you know. Deliverable: FIT / CANNOT FIT. 4. **Self-deception log (8 min).** One time you nodded at a fluent number. What costume did the work — smile, speed, digits, screen, green check? Deliverable: one named costume.

Reflect

Who should say the catch — you, the teacher, or the app — and what is stolen if someone else says it for you?

Chapter 4: Two Paths, One Quantity



Illustration for Chapter 4

A quantity does not change because you computed it differently. If two honest paths disagree, you found a leak. A tool that gives you a third number without looking at both paths is hiding the leak.

Little counted by ones then by twos. You can do a second method: mental ballpark versus written algorithm; unit analysis versus calculator; area as length times width versus counting squares; expand $3(x+4)$ by hand and by CAS. If they mismatch, the work is to find the skip — not to average, not to ask a phone which vibe it likes.

Scientists say measure twice. Programmers write a test. Accountants reconcile. This is the same civic habit: two walks, one world. Agreement is a check. Disagreement is the lesson. Do not smooth it away to keep the peace. Averaging 150 minutes and 15 minutes does not produce truth. It hides a dropped zero.

A model is not a tie-breaker for work you did not show. After both paths exist, you may ask whether grouping, converting, or distributing could skip a term. You may not ask “what is the number?” and call the fight over.

Which path felt safer, and why? Slow paths leak less and take time. Fast paths leak more and feel like talent. Talent that cannot be checked is a costume. The safer path is the one you can point at.

The scholarly neighbor here is replication. One output is one path, and it is not even yours if you cannot walk it. A second model run is not a second measurement if both runs are the same fluent engine. Two instruments that share a failure mode will agree and still be wrong. That is why the world — the pile, the drawing, the receipt — stays in view. The world is not a third model.

Worked example: 2.5 hours to minutes. Path A: 2 hours is 120 minutes; half an hour is 30; 150. Path B: 2.5×60 . If a calculator says 15, you have a unit leak. Do not average 150 and 15. Averaging hides the skip. Second pair: area of a 3-by-4 rectangle by counting squares and by 3×4 . If they fight, the grid was miscounted or the formula was applied to perimeter. The drawing is still there. A model that prints 14 without looking at both paths is not a second measurement.

Little sitting *Count It Two Ways*: if two honest paths disagree, do not ask a tool to pick a winner until both paths are on the table. A mismatch is the lesson. Do not smooth it away. A pile does not change because you counted it differently. If your two counts disagree, one path skipped or doubled. That is good news. You found a leak. Science transfer: measure twice. Integrity: do not let a model pick a winner between two unmarked guesses.

Accountants reconcile because humans skip. Labs repeat because instruments drift. You are not being asked to distrust everything. You are being asked not to treat one fluent output as a second measurement. Which path felt safer, and why? Slow paths leak less. Fast paths feel like talent. Talent that cannot be checked is a costume.

A statistical neighbor, named without pretending this is a statistics course: agreement is not validity. Two wrong thermometers in the same room will agree. Two solvers trained on overlapping text will agree. Validity is a relationship to the world. The world is the judge the sitting keeps on the table. When the pile is a word-problem, the “world” is the units and the story. When the pile is a lab, the world is the data you took. When the pile is a headline, the world is a source you can name. If you cannot name the world, you do not have two paths. You have two rumors.

The Big Idea

Matching paths are a check. A tool is not a tie-breaker for unmarked work. Averaging two disagreeing answers hides a leak.

Practice

1. **Two methods (18 min).** One quantity. Two methods on paper. No tool until both exist. If they match, write why that is not luck. If they fight, find the skip. Deliverable: both paths and the leak or the reason they should agree. 2. **Do not average (8 min).** Invent or recall two disagreeing answers. Show what averaging would hide. Deliverable: the hidden skip. 3. **CAS as second path, not first (12 min).** Hand path first. CAS second. Reconcile. Deliverable: a one-paragraph reconciliation. 4. **World still there (5 min).** Name the pile, drawing, or receipt that would judge a fight if both paths and the model disagreed. Deliverable: the named judge.

Reflect

Is a model allowed to pick a winner between two blank guesses? If you said no, where did you let it do that anyway?

Chapter 5: About, Not Fake Exact



Illustration for Chapter 5

Sometimes the honest number is “about twelve thousand,” not a costume 11,847.

Exact comes after a path. About comes first. Some quantities cannot be finished in one sitting: beans in a jar, cars in a lot, a national statistic in a headline. The work is a

responsible about, then a sample, then a scale you can say in plain words. Fermi again: a chain of abouts is more honest than a fake census.

A tool loves exact. Exact is a costume if nobody counted. Do not pretend a model counted the jar. Do not copy a model's exact count of an uncounted country onto the page. About is not laziness. Exact without a path is the laziness.

Headlines love exact. "47" looks more true than "about fifty." Size comes first. If a phone says exactly 47 for a pile nobody emptied, that 47 is less honest than your about. It borrowed finished work. Extra digits are the tell.

Civics: a budget line, a crowd size, a "percent of Americans." If you cannot name the size — tens, thousands, millions — you are not ready to share the post. Sharing a fluent exact you did not earn is how rumors get typeset.

The objection: science is precision; "about" is for children. Answer: precision is earned by a path and an error story. A lab reports 11.847 ± 0.02 because someone measured. A model that prints 11.847 for an uncounted jar has stolen the lab's costume. Gigerenzer's risk literacy is the same point with percents: a precise-looking risk with no base rate is not more scientific than "about one in a hundred of a group you can name."

If they say "about isn't real math," the sitting's reply still holds: about is the size; exact is the path finished; we need both; we do not skip to a digit we did not earn.

Little sitting *About, Not Exact*: a jar of beans, a bookshelf, cars in a lot — something you cannot finish in ten seconds. Responsible about, then a sample, then scale in plain words. No formula required at five. At fifteen you may scale more carefully, still without pretending a model emptied the jar. The guide may not emit a precise total as if it had counted. About is a grown-up word. A tool loves exact. Exact is a costume if nobody counted.

Gigerenzer's numeracy work is useful here even if you never read him. A risk of "0.14%" without a denominator is not more scientific than "about one in seven hundred of this group." The extra digits borrowed a lab coat. Crowd sizes in journalism are a known swamp: organizers' numbers, police numbers, and model-generated numbers can differ by factors, not rounding. Your honest sentence is the order of magnitude and the source. Size first. Costume exact last.

Fermi's piano tuners remain famous because the method is public. You can argue with a chain. You cannot argue with 1,247. The abuse of Fermi is treating the chain as a finding you can publish as fact. The use of Fermi is noticing that a later "exact" cannot fit the chain unless a link was badly wrong. That noticing is the alarm. Policy that uses napkin

arithmetic as if it were a census is the abuse. This booklet is on the side of the noticing, against the abuse, and honest that both exist.

If you share a number, you are making a civic claim. Sharing is not morally free. A fluent exact you did not earn, passed along, is now yours. The share test from practice is the whole ethic in one line: can you name the size class? If not, do not share.

The Big Idea

About is a number with a size. Fake exact is a rumor with extra digits. Precision without a path is a costume of science.

Practice

1. **Jar or lot (15 min).** Write about-how-many with a unit, tool closed. Count or check one sample. Scale in plain words. Then ask: would a phone's extra digits be more honest? Deliverable: about, sample, scale. 2. **Headline rewrite (12 min).** Take one precise headline number. Rewrite it as order of magnitude plus source. Mark what you do not know. Deliverable: the rewrite. 3. **Error story (10 min).** For one exact number you actually earned this week (a count, a lab, a receipt), write the path that earned it. Deliverable: path → exact. 4. **Share test (5 min).** Before you share a number, can you name the size class? If not, do not share. Deliverable: SHARE / DON'T and why.

Reflect

Would “exactly 47” from a phone be more honest than our about — and if not, why do extra digits still feel like science?

Chapter 6: The Tool as Reviewer, Not Solver

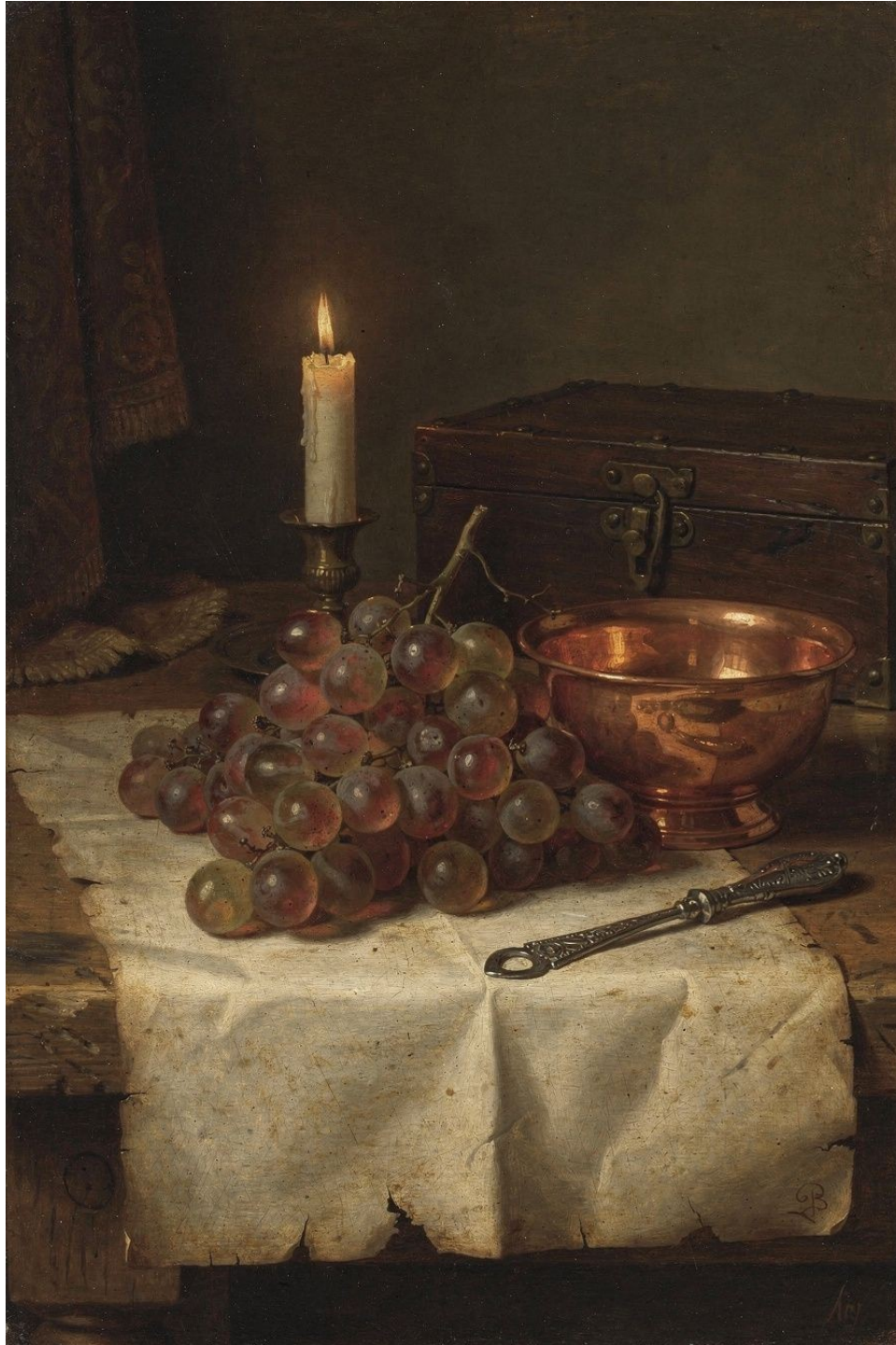


Illustration for Chapter 6

Guess. Path. Then the tool. The world you can still see is the judge.

This is the capstone, not a calculator lesson. Ballpark written. Method on the page. Then — only then — a tool may be asked whether a step is legal, a unit is wrong, a skip is likely. If it

answers with a total, close it. If it disagrees with the path you can still reproduce, believe the path until you find the leak — in you or in it.

Both leaks are possible. You skip. Tools drift. Sign errors. Training data. A confident paragraph. Neither is solved by nodding.

If the tool gives the total, that turn is a miss. Start the check again with a method question. A copied total is not a sitting. “The tool already knows” is not an argument. Knowing is not the same as you having done it. The tool can wait.

In the real world we use tools. Yes. In the real world someone still has to know when the tool is insane. That someone is the person who can still walk the path. Airplanes have instruments. Pilots still look out the window. Your math is not less serious than weather.

The Hour books remain a different product. They will ask you to do more math than this booklet. Keep this alarm when those pages get hard. Keep it when a phone is faster than your pencil. Estimate before the oracle. Path before the prize.

Miss conditions: no ballpark; no path; tool first; copied total; reverse-engineered guess; someone else announcing the catch. Any one miss and the capstone did not happen. You practiced fetching.

College applications, take-homes, and lab reports are in view. Default human path on work that wears your name. Cite influence if a tool changed a step. No model-written results. If you didn’t measure it, you don’t report it. Medical and legal claims: professionals and primary sources, not a chat.

Little capstone *Catch the Machine*: guess, path, then a tool. If the tool’s number fights your path, keep the path until you find the leak — in you or in it. Objects on the table. Guess written. Marks on paper. Then a parent-approved tool may be asked whether the method skips or doubles. If it answers with a total, close it. If it disagrees with the pile, believe the pile first. Five to ten: grown-up present. High school: you are present. The tool is never first hand. If the tool gives the total, that turn is a miss. The pile is the judge. Maybe you skipped. Maybe the tool drifted. Both are possible. Neither is solved by nodding.

The emerging sitting’s blank board is the adult of that capstone. Close the tool. Reproduce a key step. That second paragraph, that second line of algebra, is the beginning of a path that is yours. Voice is what remains when the chat is closed. Path is what remains when Photomath is closed. If you could not write the next paragraph in an exam hall, it is not your voice yet. If you could not write the next step on a blank board, it is not your math yet.

House rule, four lines, worth repeating until it is boring: TRY FIRST. PATH ON PAPER. TOOL REVIEWS. PILE JUDGES. Boring is how a rule survives 9 p.m. Kindness that pastes the

assignment into a model steals the path. Love is not a ghost. Sit on your hands if you are the parent in the room. Ask a question. Do not improve their algebra into yours.

The Hour books will go further into content. This spine is the alarm that keeps those Hours from becoming a paste. Estimate before the oracle. Path before the prize. The world on the table beats a fluent answer. Dinner: did a tool try to go first today — and what did we do? Ask it on days this booklet is closed. The house rule does not require a PDF. It requires a refusal.

The Big Idea

The tool is a reviewer. The path is the work. The world is the judge. Reproducibility is not optional because the instrument is fluent.

Practice

1. **Full three-beat (20 min)**. One problem from this week's actual math. Ballpark with unit. Path on paper. Grown-up or peer hears the path out loud. Only then a tool: did we skip, double, or break a unit? If it says the total, close it. Deliverable: the three-beat, marked pass/fail on each beat. 2. **Fight protocol (10 min)**. If tool and path fight, write the leak hunt before you believe either. Deliverable: LEAK IN ME / LEAK IN IT / NOT YET KNOWN. 3. **Closed-chat rewrite (15 min)**. After any tool use, close it and rewrite the key step from memory. Keep the one you could reproduce. Deliverable: the closed rewrite. 4. **House rule card (8 min)**. Write four lines: TRY FIRST; PATH ON PAPER; TOOL REVIEWS; PILE JUDGES. Put it where homework happens. Deliverable: the card.

Research Prompt

Read Skemp, "Relational Understanding and Instrumental Understanding," *Mathematics Teaching* 77 (1976) — it is short — and write one page: where your last week of math homework was instrumental, where it was relational, and where a solver collapsed the distinction. Do not ask a model to write the page. You may ask it for a bibliographic pointer. Cite Skemp. Cite your own homework.

Six Cases

These are not extra sittings. They are the six beats under pressure. Write in the margin whose leak it was — yours or the tool's — before you believe a digit.

Case 1 — The tip. Receipt \$38.40. Someone says put twenty percent on it. Twenty percent of forty is eight; of thirty-eight is a little under eight. About eight dollars. About forty-six total. If a phone says \$12.17, the ballpark rings. Twelve cannot be twenty percent of thirty-eight unless the custom of the house is not twenty. If a phone says \$46.08, the ballpark is quiet

enough to let the path continue: 0.20×38.40 . Quiet is not permission to skip the path. It is the alarm not ringing. Units: dollars, not percents mixed into dollars without saying so. A fluent 20 with no dollar sign is a unit costume. Reverse-engineering “about eight” after seeing \$7.68 is theater. Start again.

Case 2 — Photomath as first hand. A linear equation from the book. The camera solves it before a pencil moves. The boxed x looks like finished work. It is a prize. Reproduce the first legal step on a blank card: what did you do to both sides, and why? If you cannot say it, you did not do the math. The camera did. Second try: pencil first, full path, then the camera as a hostile reviewer of a photographed page. Ask only: is this step legal? If the app offers a full solution, that turn is a miss. Close it. Cite. If the app changed a step, write that down. If you cannot cite it, you should not have used it. Accommodations — extra time, a reader — stay named as access. They are not a license to submit a camera’s path as yours. Skemp would call the camera-first version outsourced, not even instrumental.

Case 3 — A headline digit. A post says 47,000 people were in a space you have stood in. You know the space holds hundreds on a crowded day, not tens of thousands. The ballpark is hundreds. Forty-seven thousand cannot fit. You do not need a fact-check site to say the size is wild. You may still check after. The check is second. The alarm is first. If the post says 47.382% of a group and gives no group size, you have extra digits and no base. Extra digits are a costume of measurement. Ask for the denominator before you share. Sharing is a civic act. A fluent wrong that you pass along is now yours. Little’s grown-up lie on purpose is still the drill. Someone should still lie on purpose at the table: “the drive is fourteen hours” when last week it was forty minutes. You must say it cannot be.

Case 4 — Minutes and hours. 2.5 hours to minutes. Path A: 2 hours is 120 minutes; half an hour is 30; 150. Path B: 2.5×60 . If a calculator says 15, you have a unit leak. Do not average 150 and 15. Averaging hides the skip. Second pair: area of a 3-by-4 rectangle by counting squares and by 3×4 . If they fight, the grid was miscounted or the formula was applied to perimeter. The drawing is still there. A model that prints 14 without looking at both paths is not a second measurement. Science class will call this measure twice. Keep the name if it helps. The habit is older than the class. Two solvers that agree are not two measurements if they share a failure mode.

Case 5 — The jar and the lot. Beans in a jar. About: look, say “about eighty,” write it. Sample: one handful is 19. About four handfuls that size? About eighty still holds. A phone says 83. The 83 is not more honest unless someone emptied the jar. Emptying is exact. A model’s 83 is a costume. Cars in a school lot from a window. About thirty. Count one row: eight. About four rows: about thirty-two. Still about thirty. A map app’s “parking 27” is a different claim — it may be using another source. It is not your looking. Label it as the app. Do not write 27 as if you counted 27. Headline: “exactly 19,447 jobs.” Unless you have the

agency table, your honest sentence is the order of magnitude and the source. Size first. Costume exact last. Gigerenzer: a precise-looking percent with no base is not more scientific than an honest about.

Case 6 — The full three-beat. One problem from this week’s actual math. Write a ballpark with a unit. Write the path. Keep the paper in view. A grown-up or peer hears the path out loud. Only then a tool: did we skip, double, or break a unit? If it says the total, close it. If it fights the path, believe the path until the leak is found — sign error, dropped unit, or a fluent hallucination. Miss conditions: no ballpark; no path; tool first; copied total; reverse-engineered guess; someone else announcing the catch. Any one miss and the capstone did not happen. You practiced fetching. The pile is still the judge. For you the pile is the problem in the book, the receipt, the map, the lot, the gym. The world you can still see beats a fluent digit. Guess. Path. Then the tool. If the network died, which step could you still walk?

Case 7 — Take-home and lab. A take-home set that a model can finish in four minutes is a test of whether you still have a path, not a test of whether you can find the model. Default human path. Cite if a tool changed a step. A lab notebook that contains “Grok said the rate should be...” as if that were data is a category error. Mechanism stories are cheap. Measurement is expensive. Know which you are doing. Nature does not autocomplete. You have to go there. The science sitting *The Universe Is Not a Chatbot* is the sibling; do not mash it into this PDF. Keep the ethic: no model-written results.

Case 8 — Money in the news. A program “costs \$4 million.” Last week’s article was hundreds of billions. You have a unit problem or a lie. You cannot catch it without a ballpark. Thousand, million, billion are different rooms. If you cannot name the room, you are not ready to share the post. This is Gigerenzer’s civic numeracy without requiring his book on the table. Size first.

Debates You Should Be Able to Hold

A high booklet that only cheers for “try first” is a poster. These are the fights. Take the other side at full strength, then say where you stand.

1. “Professionals use CAS, so withholding it is museum pedagogy.” Full strength: working mathematicians, engineers, and analysts use computer algebra, notebooks, and solvers every day. Training people on pencil-only is training them for a job that no longer exists. Reply: professionals still have to know when the instrument is insane, defend a step, and reproduce a result without the original session. High school is building that defense-of-step muscle. The exam hall is a reproducibility condition, not a museum. A junior who cannot tell a wild output from a live one is not “using modern tools.” They are a liability with a login. Stand: tools as reviewers after a path exists. Not first hand on work that wears your name.

2. “Fermi estimates license sloppy policy.” Full strength: napkin arithmetic has been used to sell wars, budgets, and products. “About” becomes a way to skip measurement. Reply: a Fermi chain is a prior. Priors exist so you can notice when a later number cannot fit. If you treat the prior as the finding, you have cargo-culted Fermi. The abuse is real. The use is the alarm. Stand: public chain, named uncertainties, never published as a census. Argue with the links. Do not argue with a costume 1,247.

3. “Worked examples mean I should watch the solver.” Full strength: Sweller’s research says novices learn from worked examples; struggling without them can overload working memory. Reply: a worked example you cannot reconstruct is a video of someone else walking. Watch, then close, then walk. If the example never closes, you did not study. You consumed fluency. Stand: one worked example, then a near problem with the example closed. The sitting is the walk.

4. “Math anxiety means they should use the tool so they don’t freeze.” Full strength: anxiety is real. Shame is a real wrecker. A student who panics at a blank page may need access: extra time, a reader, a quieter room, a smaller pile. Reply: access is not a ghost. A model that does the math so nobody has to feel the freeze deletes the person you were trying to protect. Little’s rule still holds: do not shame a wild guess; wild is data. Stand: accommodations named as access; ballpark still first; tool still second. Anxiety is a reason to change the conditions, not to outsource belief.

5. “The teacher can’t tell, so the ethic is unenforceable.” Full strength: detection is imperfect. Fluency is cheap. Schools will fail to catch a fraction of ghosted work. Reply: you can tell. A later teacher can tell. A blank exam hall will tell. A boss who needs a number defended will tell. Enforceability is not the same as the ethic. English already lives this. Math is not exempt because the symbols look objective. Stand: cite; reproduce; no application ghosts. The house rule is for you, not for the detector.

6. “Everyone will use it in college anyway.” Full strength: they will. Policies are uneven. Some courses will allow solvers. Reply: then the ones who can still walk the path will be the ones worth hiring and trusting. Be those. College that outsources all derivation will still have moments — labs, interviews, accidents — where a fluent wrong has to be caught by a human with a ballpark. Stand: default human path on work that wears your name. Tools review. The world judges.

7. “Precision is science; about is for children.” Full strength: laboratories report significant figures for a reason. Engineering tolerances are not vibes. Reply: precision is earned by a path and an error story. A lab’s 11.847 ± 0.02 is not the same object as a model’s 11.847 for an uncounted jar. The second stole the lab coat. Gigerenzer: a precise percent

with no base is not more scientific than “about one in a hundred of a group you can name.”
Stand: exact after a path; about when the path is not finished; never costume exact.

8. “Cargo cult is a metaphor too cute for homework.” Full strength: Feynman was talking to scientists about experiments that imitated the forms of investigation. Homework is not a radar station. Reply: the form he named is imitation-without-honesty. A boxed answer with extra digits and a therefore-paragraph can be that imitation. The plane is a path you can walk when the chat is closed. If the path is not there, the look of math did not cause understanding. Stand: use the metaphor once, then do the sitting. Do not live in the metaphor.

Hold these eight without needing to win a dinner. The deliverable is that you can steel-man the other side in four sentences, then state the standing rule in two.

How to Run the Six Sitzings

The booklet is not a replacement for the live sittings. It is a spine you can hold when the site is closed. Run them in order. Do not skip sitting 3 because a lie on purpose feels unkind.

Sitting 1 — Guess, Then Count / Estimate Before the Oracle. Objects or a real quantity. Write the guess or ballpark with a unit. Then count or compute. Then, if you want, a tool may check the method. Not the total. If they skip the guess, start again. Do not shame a wild guess. Wild is data.

Sitting 2 — The Path on Paper. Marks or steps before any model. The guide may not give the count. The tool is closed while they count. It may see the path after. If they ask how many, send them back to the paper. Scribe if they cannot write; do not ghost.

Sitting 3 — Catch a Wild Number. A grown-up supplies one confident wrong total. The student must say the catch. The device is not first. Repeat on a miss. Optional later: a tool question they already know, to see if the alarm still rings.

Sitting 4 — Two Ways. Same quantity, two paths, both on the table before any tool. A mismatch is the lesson. Do not average. Do not let a model pick a winner between unmarked guesses.

Sitting 5 — About, Not Exact. A pile too big to finish fast. Responsible about, then a sample, then plain-word scale. No fake exact from a model that did not count the jar.

Sitting 6 — Catch the Machine. The three-beat as capstone. Guess. Path. Tool as reviewer. If the tool gives a total, miss, start the check again. The pile — or the problem still in view — is the judge.

Pólya's four phases in *How to Solve It* map onto this arc without forcing a costume. Understanding the problem is sitting 1's unknown, data, and condition — the ballpark. Devising a plan is sitting 2's path. Carrying out the plan is the count or the algorithm, still on paper. Looking back is sittings 3–6: catch a wild, measure twice, honest about, tool as reviewer. You do not need to lecture the four phases to a twelve-year-old. You need to run them.

If homework is due in twenty minutes, you may still run a shorter three-beat on one problem instead of a camera on five. One honest path is more math than five prizes. The Hour books can take the rest of the week. This spine is the alarm that keeps those Hours from becoming a paste.

A Week of Ballparks

Run one small three-beat a day for a week. None of these need a special kit. All of them need a closed tool until the ballpark and the path exist.

Monday — Tip. A real receipt. Ten percent is moving the point. Fifteen is that plus half of it. Twenty is double ten. Write the about. Then compute. Then a tool may check the multiplication. Medical bills and alcohol rules are grown-up policy; this is the arithmetic of a sandwich.

Tuesday — Time. A drive or a walk you have actually taken. About how many minutes. Then a clock or a map as check, not a model first. If the app says fourteen hours for a forty-minute town, catch it. Units: hours are not minutes.

Wednesday — Homework. One problem from the book that is already assigned. Ballpark with unit. Path on paper. Tool as reviewer of steps. If the app solves it in the camera, that day is a miss even if the boxed number is right.

Thursday — Two paths. Same quantity two ways. Ones and groups, or algorithm and ballpark, or minutes two ways. If they fight, find the skip. Do not average. Do not ask a phone which vibe it likes.

Friday — About. A jar, a shelf, a lot, a headline. Write about. Sample. Scale in plain words. Refuse a costume exact.

Saturday — Catch. A grown-up lies on purpose, kind and sure. You say it cannot be. Name the costume: smile, speed, digits, screen.

Sunday — Look back. Pólya's fourth phase. What leaked this week? Where did a tool go first? Where did a path get stolen by kindness? Write three lines. That is the week. The sittings are still there on Monday.

This week is not a new sitting unit. William does not need to draft it. It is practice of the six that already exist. If a day is chaos, skip the calendar and keep the three-beat on whatever number entered the house.

Two Kinds of Knowing

Richard Skemp, writing in 1976 for mathematics teachers, distinguished *instrumental* understanding (rules without reasons) from *relational* understanding (knowing what to do and why). A camera that boxes an x is instrumental in the extreme: you may not even hold the rule. This booklet is not anti-rule. Algorithms are paths you can walk. It is anti-prize-without-path.

Instrumental knowing feels fast. Relational knowing survives a blank board. Pólya's questions are a relational drill. Photomath-first is an instrumental drill that pretends to be help. If you can only say "the app did it," you have neither. Skemp's warning was about classrooms that reward speed. Talking tools reward fluency. Fluency is not relational knowing.

You do not need Skemp's paper on the table. You need the test he implies: can you walk the path when the instrument is closed? If yes, the instrument may come back as a reviewer. If no, you are renting.

Gigerenzer's work on numeracy and risk is the civic twin. People who cannot estimate a size will believe a fluent percent. "47%" with no base is not information. "About one in two" of a group you can name is information you can argue with. Headlines prefer the costume. This spine prefers the size.

Sweller's worked-example research says novices learn from worked examples. That is not permission to skip doing one. A worked example you cannot reconstruct is a video of someone else walking. Watch, then walk. The sitting is the walk.

None of these names are required on a test. They are here so a grown-up can see that "guess then count" is not a cute slogan. It is the same argument math education has been having for fifty years, now with a model that will happily be the whole class.

If They Say

These lines are already in the sittings. Keep them. Do not improve them into a lecture.

"Guessing is stupid if the tool knows." The tool does not know if it drifted. Your guess is the alarm. Bad at math is not a reason to outsource belief.

"The calculator is the method." The calculator is an instrument. The method is the sentences you could say without it.

“Just tell me how many.” That is the prize. The path is the work. Show the marks.

“The tool wouldn’t be wrong.” Tools drift. Voices sound sure. Your guess is how you notice.

“Let’s just ask the tool which is right.” The pile is still here. The marks are still here. Find the skip first.

“About isn’t real math.” About is the size. Exact is the path finished. We need both. We do not skip to a digit we did not earn.

“The tool already knows how many.” Then it can wait. We still guess. We still mark. Knowing is not the same as us having done it.

“In the real world we use tools.” In the real world someone still has to know when the tool is insane. That someone is the person who can still walk the path.

“The teacher can’t tell.” You can tell. So can a later teacher, a boss, and the blank page in an exam hall.

“Everyone will use it anyway.” Then the ones who can still walk the path will be the ones worth trusting. Be those.

Hard edges, collected: do not shame a wild guess; the guide may not give the count; no live timed test as a demo if it tempts cheating; units matter; CAS and Photomath are in scope; accommodations are access, not a ghost; cite the tool if it changed a step; medical dosing is a professional; do not let a model be the first check on a news statistic; if the tool gives the total, that turn is a miss.

Dinner questions, collected: What did we guess before we knew? What path did we draw before we believed a number? What wrong number did we catch — and how did we know? When did two ways disagree — and what did we find? Where did about tell the truth better than a fake exact? Did a tool try to go first today — and what did we do?

If you only remember one list, remember the dinners. They are the house rule in six lines.

Transfer Map

The sittings already name transfers. Keep them honest. Do not invent new subjects to look busy.

| From this spine | Into | | --- | --- | | Ballpark before the oracle | Science: prediction before measurement | | Ballpark before the oracle | Civics: budgets and headlines | | Path on paper | Thinking: claim and check | | Path on paper | AI: ask about the path; refuse a bare digit | | Two paths | Science: measure twice | | About, not exact | Civics: headlines

love exact; size comes first | | Catch the machine | AI: estimate before the oracle; path before the prize | | Catch the machine | Science: the world on the table beats a fluent answer |

English has a twin sitting: *You Write It*. Code has a twin: explain every line. History has a twin: source before summary. Those are other units. Do not mash them into this PDF. This PDF is number.

A Note on Kindness

Kindness that says the total steals the path. Kindness that announces the catch steals the alarm. Kindness that pastes the assignment into a model steals the voice in English and the path in math. Love is not a ghost. Sit on your hands. Ask a question. Scribe exact words if they cannot write. Do not improve them into you.

Five to ten still need a grown-up in the room. Eleven to fourteen can hold the pencil. You still hold the closed-tool rule. If you fail, say so. That honesty is the sitting. Shame is not the sitting. Wild guesses are data. Missed catches are data. Reverse-engineered ballparks are theater; start again.

What You Owe a Number That Wears Your Name

A transcript is a claim about a mind. So is a lab notebook. So is a news post with your account under it. The number is not the only thing that wears your name. The path does. If the path is a rental, the name is a costume.

This is not a purity rule against instruments. It is a reproducibility rule. Could you walk this on a blank board? Could you tell a peer where it is shaky? Could you notice a fluent 400 against a ballpark of 10? If yes, the instrument may come back as a reviewer. If no, you are not done, even if the boxed answer is right.

Right answers with missing paths are how people learn that math is a performance. Performance is what cargo cults imitate. Understanding is what survives a closed laptop. Pólya wanted you to have questions you could ask when stuck. Feynman wanted you not to fool yourself. Skemp wanted reasons, not only rules. Gigerenzer wanted sizes you could argue with. This academy sitting wanted grapes on a table before a talking tool. The through-line is one law: the world first, then the path, then the instrument, then the digit.

Keep the law when the Hour books get hard. Keep it when a phone is faster than a pencil. Keep it when a parent is kind at 9 p.m. Kindness that pastes is not kindness in this house. Sit on your hands. Ask a question. The pile is still the judge.

The six slugs remain `estimate-first`, `path-on-paper`, `check-the-wild-answer`, `same-quantity-two-ways`, `about-not-exact`, `catch-the-machine`. Do not invent a seventh sitting to

look busy. Do not unfreeze Bonaventure, Thérèse, or Stoicism because this spine exists. Do not open *Measure Twice* until this pack is actually live. WIP max 1. Payment pulled. All booklets free. No purchase CTA. Liam maps when live PDF+`.md` pass, not when a draft is on disk.

If this file is under 60 KB it is not ready to convert. Clare taught the under-floor lesson. Bonaventure's live `.md` under floor is the other warning. High's floor is sixty. Do not ship a 25 KB essay and call it the high band because the chapters exist. Structure is not gold. Bytes of sourced argument are the floor, then unique art, then multi-MB PDFs, then live curl.

Medical dosing is never this booklet's example for a reason. A model that is fluent about milligrams is still not a clinician. If a household number is a dose, the path is a professional and a labeled bottle, not a chat. That hard edge is in the adult sitting too. Copy it here so a clever student does not "practice Fermi" on medicine.

News statistics: do not let a model be the first check. Official pages, receipts, maps, agency tables. Then, if you want, a model may list ways a headline number might be off. That list is not proof. It is a set of questions. Proof is the source.

If you only remember four words from this band, remember these: path over prize. Then go back to the grapes. The grapes were never baby math. They were the whole law at a scale you could hold.

For the Grown-Up Reader

High-band booklet for *Number in the Age of AI*. Little 6/6 sittings are live. This prose follows those six as academic integrity. It is not *Mathematics for Homeschooling*. It is not a philosopher booklet. It is not gold until live `/downloads/number-in-the-age-of-ai-high.{pdf,md}` return 200 at floor.

Sources: Pólya, *How to Solve It* (Princeton, 1945). Feynman, "Cargo Cult Science," Caltech 1974. Skemp, *Mathematics Teaching* 77 (1976). Fermi named for order-of-magnitude practice; this band does not quote a primary Fermi paper. Gigerenzer named for numeracy and base rates. Sweller named for worked examples. NCTM named for estimation as a standard. Academy lines are labeled as sittings.

Hard edges: no live timed test as a demo if it tempts cheating; units matter; CAS and Photomath in scope; accommodations are access; cite the tool if it changed a step; medical dosing is a professional; do not let a model be the first check on a news statistic; if the tool gives the total, miss.

The six slugs: `estimate-first`, `path-on-paper`, `check-the-wild-answer`, `same-quantity-two-ways`, `about-not-exact`, `catch-the-machine`. Liam maps when the pack is actually live. Do

not invent young sittings 2–6. Sitting 1 already exists in young. William drafts only if a sitting is missing, which it is not.

About WisdomForge

WisdomForge writes the same idea four times: elementary, middle, high, and adult. The facts do not change. The sentences do. This is the high band of *Number in the Age of AI*. Sittings stay live. This booklet is the download. Payment is pulled. There is no purchase CTA. All booklets are free. The exam hall is a reproducibility condition, not a museum.

Sources

Primary. George Pólya, *How to Solve It* (Princeton University Press, 1945). Richard Feynman, “Cargo Cult Science,” Caltech commencement, 1974. WisdomForge sittings, *Number in the Age of AI* (‘estimate-first’ through ‘catch-the-machine’).

Secondary. Richard Skemp, “Relational Understanding and Instrumental Understanding,” *Mathematics Teaching* 77 (1976). Gerd Gigerenzer on numeracy and risk literacy. John Sweller on worked examples. NCTM estimation standards. Enrico Fermi’s order-of-magnitude practice as a public chain, not a fake census.

The pile is still the judge. Guess. Path. Then the tool. Digit last. Keep the alarm. The world first.